TPM4-related Platelet Disorder (MONDO:0100487) — Comprehensive Disease Characteristics Report

Causal gene: *TPM4* (tropomyosin 4; HGNC:12013; encoding the cytoskeletal/non-muscle tropomyosin isoform Tpm4.2). This report concerns the autosomal dominant inherited macrothrombocytopenia caused by heterozygous variants in **TPM4**, not the muscle tropomyosinopathies caused by the related genes *TPM1*, *TPM2*, or *TPM3*. All content below pertains to the TPM4 platelet disorder.

Summary

TPM4-related Platelet Disorder is a rare, **autosomal dominant inherited macrothrombocytopenia** caused by heterozygous loss-of-function (truncating) variants and rare deleterious missense variants in **TPM4**, the gene encoding the cytoskeletal (non-muscle) tropomyosin isoform **Tpm4.2**. The disorder was first defined as a Mendelian entity in 2017, when two unrelated families from the BRIDGE Bleeding and Platelet Disorders (BPD) collection were found to carry TPM4 truncating variants that segregated with low platelet counts and enlarged platelets (PMID: 28134622). The gene had previously been flagged through genome-wide association studies of platelet count and mean platelet volume in ~70,000 individuals, which was the lead that prompted the Mendelian search.

Mechanistically, TPM4 behaves as a **haploinsufficient** gene. A heterozygous truncating or deleterious missense variant reduces the amount of functional Tpm4.2 available to specify and stabilize the megakaryocyte actin cytoskeleton. Because Tpm4.2 binds F-actin with high affinity, competes with α-actinin, and regulates myosin and ADF/cofilin activity, its reduction destabilizes and mis-specifies actin–tropomyosin filaments. This impairs the **terminal stages of platelet production** (proplatelet formation and platelet release from megakaryocytes), yielding **fewer but larger platelets** — the defining macrothrombocytopenia — with a mild additional effect on platelet function. The mechanism is conserved: ENU-induced missense mutations and targeted inactivation of *Tpm4* in mice reproduce a gene-dosage-dependent macrothrombocytopenia with otherwise normal blood counts (PMID: 28134622).

Clinically, the phenotype is **mild and lifelong (non-progressive)**: mild-to-moderate thrombocytopenia, a high proportion of enlarged platelets on the blood film, and a mild mucocutaneous bleeding tendency. Standard platelet aggregation and granule secretion tests are frequently normal, so the condition is easily mistaken for immune thrombocytopenia (ITP) and **requires genetic testing (whole-exome sequencing or inherited-platelet-disorder gene panels) for a definitive diagnosis**. Missense-variant carriers (e.g., p.R182C, p.A183V) can display more prominent platelet-function defects and more significant bleeding. There is no disease-specific therapy; management is supportive (antifibrinolytics such as tranexamic acid, hormonal control of menorrhagia, desmopressin, and peri-procedural platelet support when needed).

Key Findings

Finding 1 — TPM4 truncating variants cause autosomal dominant macrothrombocytopenia

The disease-defining observation comes from Pleines et al. 2017, who identified **two unrelated families** in the BRIDGE-BPD collection carrying a TPM4 variant that truncates the TPM4 protein and **segregates with macrothrombocytopenia** — a disorder of low platelet count and enlarged platelets (PMID: 28134622). Direct quote: *"we have identified 2 unrelated families in the BRIDGE Bleeding and Platelet Disorders (BPD) collection who carry a TPM4 variant that causes truncation of the TPM4 protein and segregates with macrothrombocytopenia, a disorder characterized by low platelet count."*

Causality is reinforced by animal modeling in the same study: *"N-Ethyl-N-nitrosourea–induced (ENU-induced) missense mutations in Tpm4 or targeted inactivation of the Tpm4 locus led to gene dosage-dependent macrothrombocytopenia in mice"* — establishing both causation and a **gene-dosage (haploinsufficiency)** mechanism (PMID: 28134622).

An independent Spanish pedigree later confirmed the entity: whole-exome sequencing revealed a **novel nonsense variant c.322C>T (p.Gln108*)** in TPM4 (NM_003290.3) segregating in 5 family members with lifelong mild thrombocytopenia and enlarged platelets (PMID: 35170221). Quote: *"WES revealed a novel nonsense variant (c.322C>T; p.Gln108*) in TPM4 (NM_003290.3), the gene encoding for tropomyosin-4 (TPM4)."* Together these three pedigrees (two in 2017, one in 2022) establish truncating/nonsense TPM4 variants as the cause of dominant macrothrombocytopenia in humans.

Finding 2 — TPM4 deficiency impairs terminal platelet production and causes cytoskeletal/platelet-function defects

The core cellular defect is a failure of the **final steps of platelet manufacture**. In Pleines et al., *"Insufficient TPM4 expression in human and mouse megakaryocytes resulted in a defect in the terminal stages of platelet production and had a mild effect on platelet function"* (PMID: 28134622).

Functional studies of the p.Gln108* nonsense variant demonstrated a downstream cytoskeletal consequence: the variant *"led to impairment of platelet spreading capacity after stimulation with TRAP-6 and CRP, delocalization of TPM4 in activated platelets, and significantly reduced TPM4 levels"* (PMID: 35170221). The reduced protein level is direct evidence of haploinsufficiency, and the spreading/localization defects show impaired cytoskeletal remodeling in activated platelets.

Rare conserved **missense variants (p.R182C and p.A183V)** in 3 patients from 2 families extend the spectrum toward platelet dysfunction: these variants caused *"reduced platelet secretion, and aggregation and spreading defects suggesting that TPM4 missense variants impact platelet function and show a disordered pattern of tropo[myosin]"* (PMID: 34758189). Finally, the megakaryocyte partner protein **tropomodulin-3 (Tmod3)** associates with TM4/TPM4 and F-actin; its loss disrupts the demarcation membrane system and proplatelet formation, placing TPM4 within a defined actin-organizing module of platelet biogenesis (PMID: 25964668).

Finding 3 — TPM4 encodes a cytoskeletal (non-muscle) tropomyosin that specifies and stabilizes actin filaments

TPM4 is one of **four vertebrate tropomyosin genes (TPM1–TPM4)**, from which more than 40 isoforms arise through alternative splicing and alternative promoters (PMID: 28101865; PMID: 28216317). The relevant product in platelets/megakaryocytes is the cytoskeletal isoform **Tpm4.2**, a stress-fiber-associated tropomyosin. In vitro reconstitution showed that these cytoskeletal tropomyosins *"bound F-actin with high affinity and competed with α-actinin for actin filament binding"* (PMID: 28216317), and that they regulate myosin activity and protect filaments from ADF/cofilin-mediated severing.

At the conceptual level, *"Tropomyosin can be viewed as a universal regulator of the actin cytoskeleton that specifies actin filaments for cellular and intracellular functions"* (PMID: 28101865). In the specific context of megakaryocytes, Tmod3 *"associates with TM4 and the F-actin cytoskeleton in wild-type MKs"*, and confocal microscopy shows Tmod3, TM4, and F-actin

partially co-localizing near proplatelet buds (PMID: 25964668). This provides the molecular rationale for why reducing Tpm4.2 destabilizes the actin cytoskeleton that drives proplatelet formation.

Finding 4 — The disorder is a mild, lifelong autosomal dominant macrothrombocytopenia with low bleeding burden

Across the three reported pedigrees the clinical picture is consistent: **lifelong mild-to-moderate thrombocytopenia, a high number of enlarged platelets on blood film, and a mild mucocutaneous bleeding tendency**. In the Spanish family, *"Five family members displayed lifelong mild thrombocytopenia with a high number of enlarged platelets in blood film, and mild bleeding tendency. Patient's platelets showed normal aggregation and granule secretion response to several agonists"* (PMID: 35170221) — emphasizing that standard functional tests can be normal.

The more severe end of the spectrum is seen with missense alleles: *"rare conserved missense variants (p.R182C and p.A183V) in TPM4 encoding tro[po]myosin-4 in 3 patients"* were associated with platelet dysfunction and excessive bleeding (PMID: 34758189). The TPM4 platelet-count/volume signal was first detected by GWAS in ~70,000 individuals (PMID: 28134622), and the disorder is diagnosed within the framework of gene panels/WES for inherited platelet disorders (PMID: 19408193 for the general diagnostic paradigm).

Finding 5 — Consolidated causal chain: TPM4 haploinsufficiency → macrothrombocytopenia

Integrating all evidence yields a stepwise mechanism from mutation to clinical phenotype (detailed in the Mechanistic Model below). The conserved, nonredundant role of TPM4 across species is captured by the original paper's conclusion of *"a nonredundant role for TPM4 in platelet biogenesis in humans and mice"* (PMID: 28134622).

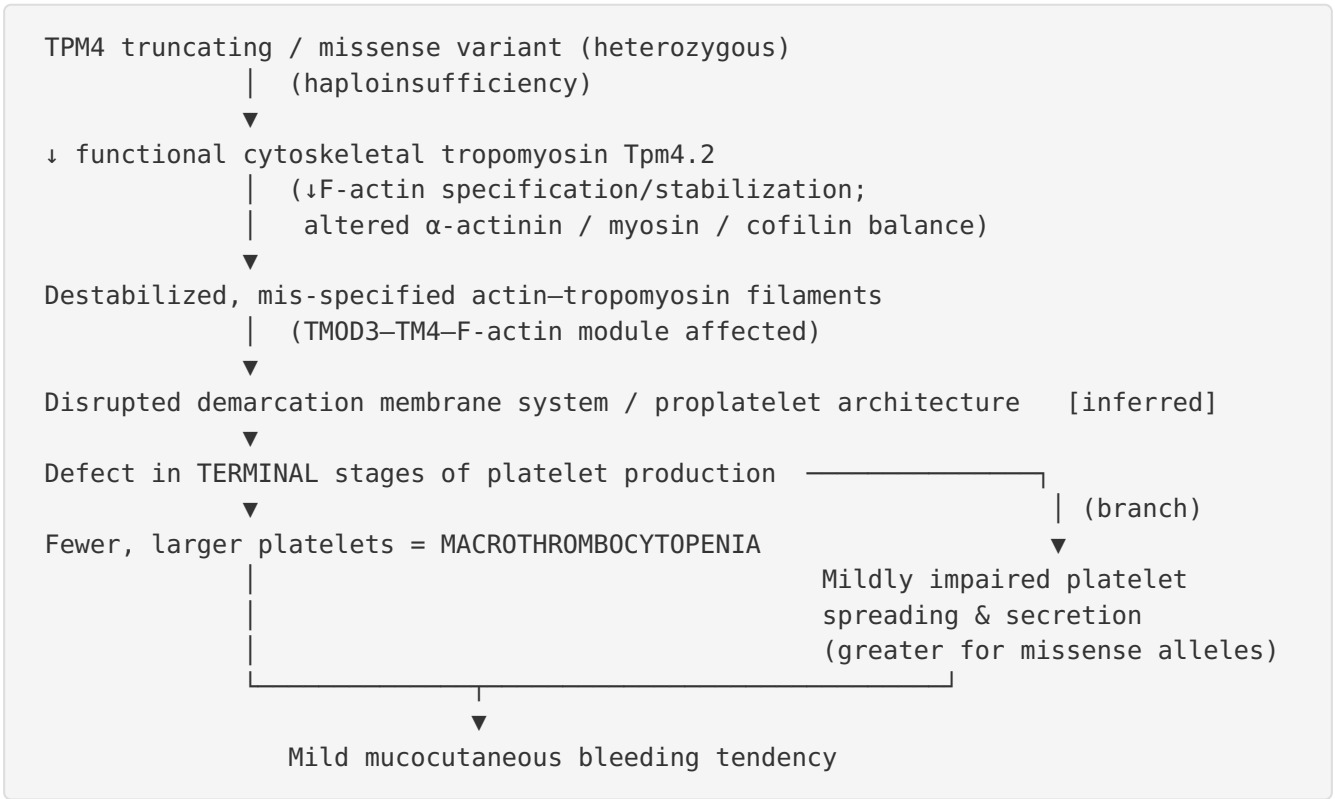
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous TPM4 truncating variant** (e.g., p.Gln108*) or deleterious missense variant (e.g., p.R182C, p.A183V) **leads to** reduced functional cytoskeletal tropomyosin Tpm4.2

(haploinsufficiency — reduced TPM4 protein directly demonstrated, PMID: 35170221; gene-dosage dependence shown in mice, PMID: 28134622).

2. Reduced Tpm4.2 **results in** destabilized and mis-specified actin–tropomyosin filaments, because Tpm4.2 normally binds F-actin with high affinity, competes with α-actinin, and regulates myosin/cofilin (in vitro; PMID: 28216317, PMID: 28101865).
3. Destabilized filaments **lead to** disrupted megakaryocyte demarcation membrane system and impaired actomyosin-driven proplatelet architecture (*inferred*, supported by the TMOD3–TM4–F-actin axis in megakaryocytes; PMID: 25964668).
4. This **results in** a defect in the terminal stages of platelet production (directly demonstrated in human and mouse megakaryocytes; PMID: 28134622).
5. The production defect **produces** fewer, larger platelets = **macrothrombocytopenia** (demonstrated).
6. **Branch:** reduced/mis-localized Tpm4.2 in circulating platelets **also leads to** mildly impaired platelet spreading and secretion — more pronounced for missense alleles (PMID: 35170221, PMID: 34758189).
7. Steps 5 and 6 together **result in** a **mild mucocutaneous bleeding tendency**.



Upstream vs downstream, cells, and biological processes

- **Upstream:** the genetic lesion and the resulting reduction in Tpm4.2 protein (molecular/protein level).
- **Midstream:** actin-cytoskeleton dysregulation within **megakaryocytes** (CL:0000556, megakaryocyte) — the cell type where platelet biogenesis occurs.
- **Downstream:** quantitative and qualitative abnormalities of circulating **platelets** (CL:0000233, platelet).

Suggested GO biological-process terms: platelet formation (GO:0030220), megakaryocyte development (GO:0035855), actin filament organization (GO:0007015), actin cytoskeleton organization (GO:0030036), regulation of actin filament depolymerization (GO:0030834). **Suggested GO molecular-function terms:** actin binding (GO:0003779), actin filament binding (GO:0051015). **Suggested GO cellular-component terms:** actin cytoskeleton (GO:0015629), stress fiber (GO:0001725), cortical actin cytoskeleton (GO:0030864). **Suggested CL terms:** megakaryocyte (CL:0000556), platelet (CL:0000233). **Suggested CHEBI/related entities:** tranexamic acid (CHEBI:48669), desmopressin (CHEBI:4450).

Section-by-Section Report

1. Disease Information

- **Overview:** A rare Mendelian **autosomal dominant macrothrombocytopenia** — reduced platelet count with abnormally large platelets — caused by TPM4 variants, typically mild and lifelong.
- **Key identifiers: MONDO:0100487**; **Gene:** TPM4 (HGNC:12013; NCBI Gene 7171; Ensembl ENSG00000167460; UniProt P67936). OMIM, Orphanet, ICD-10/11, and MeSH disease-specific identifiers were **not recorded/confirmed** in the available evidence; this is a recently delineated entity (2017) and may be captured under broad terms such as "constitutional/inherited macrothrombocytopenia." The reference transcript is **NM_003290.3**.
- **Synonyms:** TPM4-related macrothrombocytopenia; TPM4-related thrombocytopenia; tropomyosin-4-related platelet disorder. (No formally recorded synonym list available.)
- **Data source type:** Information is derived from **individual patients/families** (case series and pedigrees) combined with **aggregated population genomics** (GWAS of platelet traits) and **model-organism** data — not from a disease-level EHR registry.

2. Etiology

- **Causal factors:** Purely **genetic** — heterozygous pathogenic variants in TPM4. No environmental or infectious cause.
- **Genetic risk factors:** The causal variants themselves (truncating/nonsense and rare deleterious missense). GWAS identified TPM4 as influencing platelet count and mean platelet volume in ~70,000 individuals, the common-variant lead that preceded Mendelian confirmation (PMID: 28134622).
- **Environmental/lifestyle risk factors:** None established. Age, sex, and exposures do not cause the disorder, though they may modulate bleeding expression (e.g., menorrhagia in women, bleeding with surgery/trauma).
- **Protective factors:** None specifically identified (genetic or environmental).
- **Gene–environment interactions:** None documented. Bleeding risk is expected to rise with hemostatic challenges (surgery, trauma, antiplatelet/anticoagulant drugs), but this is a general consequence of thrombocytopenia rather than a specific GxE interaction.

3. Phenotypes

Phenotype	Type	HPO term	Onset	Severity	Course	Frequency
Thrombocytopenia (low platelet count)	Laboratory abnormality	HP:0001873	Congenital/ lifelong	Mild–moderate	Stable, non-progressive	~100% of affected
Enlarged platelets / ↑MPV	Laboratory abnormality	HP:0040310 (large platelets) / HP:0011877 (↑MPV)	Congenital/ lifelong	—	Stable	High (hallmark)
Mucocutaneous bleeding (epistaxis, bruising, menorrhagia)	Clinical sign/ symptom	HP:0000978 (bruising), HP:0000421 (epistaxis), HP:0000132 (menorrhagia), HP:0001892 (abnormal bleeding)	Lifelong	Mild (missense alleles more severe)	Episodic with challenges	Variable; often mild/ absent
Impaired platelet spreading / secretion	Laboratory abnormality (functional)	HP:0011869 (abnormal platelet function)	—	Mild; greater for missense	Stable	Variant-dependent

- **Quality-of-life impact:** Generally low; most affected individuals are minimally symptomatic. Menorrhagia and surgical/peri-procedural bleeding are the main QoL concerns in symptomatic carriers. No disease-specific EQ-5D/SF-36/PROMIS data available.
- Normal aggregation and granule secretion can occur despite the genetic defect (PMID: 35170221).

4. Genetic / Molecular Information

- **Causal gene: TPM4** (tropomyosin 4), chromosome 19p13.1; HGNC:12013; UniProt P67936; transcript **NM_003290.3**.
- **Reported pathogenic variants:**

Variant (cDNA / protein)	Type	Classification	Functional consequence	Reference
Truncating variants (2 BRIDGE-BPD families)	Truncating	Pathogenic	Loss of function / haploinsufficiency	PMID: 28134622
c.322C>T (p.Gln108*)	Nonsense	Pathogenic (segregates in 5 relatives)	↓TPM4 protein; impaired spreading	PMID: 35170221
p.R182C	Missense (conserved)	Likely pathogenic	↓secretion, aggregation/ spreading defect; excessive bleeding	PMID: 34758189
p.A183V	Missense (conserved)	Likely pathogenic	Platelet dysfunction, disordered tropomyosin pattern	PMID: 34758189

- **Allele frequency:** All reported variants are rare/private (not population-polymorphic); specific gnomAD frequencies not recorded here but expected to be absent or ultra-rare.
- **Somatic vs germline: Germline**, constitutional.
- **Functional consequences:** Truncating/nonsense = **loss of function (haploinsufficiency)**; missense = deleterious with likely dominant-negative/mis-specification of the actin–tropomyosin filament (disordered tropomyosin pattern). Gene-dosage dependence demonstrated in mice (PMID: 28134622).
- **Modifier genes:** None established; TMOD3 (tropomodulin-3) is a functional partner in megakaryocyte actin organization but not reported as a clinical modifier (PMID: 25964668).
- **Epigenetic information / chromosomal abnormalities:** None reported for this disorder.

5. Environmental Information

- **Environmental factors, lifestyle factors, infectious agents: Not applicable.** This is a monogenic germline disorder with no known environmental, lifestyle, or infectious contribution. (Hemostatic stressors influence bleeding expression but do not cause the disease.)

6. Mechanism / Pathophysiology

See the **Mechanistic Model** above for the full ordered causal chain, ASCII diagram, and ontology-term suggestions. Key points: - **Molecular pathway:** actin-cytoskeleton regulation via tropomyosin specification of actin filaments (not a classical signaling cascade such as Wnt/MAPK). Tropomyosins compete with α-actinin, regulate myosin II, and protect filaments from ADF/cofilin severing (PMID: 28216317). - **Cellular process:** defective proplatelet formation / terminal platelet production in megakaryocytes (PMID: 28134622). - **Protein dysfunction:** reduced/mis-localized Tpm4.2 (loss of function for truncating; mis-specification/possible dominant-negative for missense). - **Subcellular compartments:** actin cytoskeleton, stress fibers, demarcation membrane system of megakaryocytes. - **Immune/metabolic involvement:** none intrinsic. (Note: distinct, non-disease-causal TPM4 biology appears in cancer contexts — e.g., TPM4 modulating ferroptosis via SCD1 in gastric cancer, PMID: 40214825 — but these are unrelated to the platelet disorder.) - **Molecular profiling:** direct demonstration of reduced TPM4 protein in patient platelets (PMID: 35170221); disordered tropomyosin filament pattern with missense variants (PMID: 34758189).

7. Anatomical Structures Affected

- **Organ/system:** hematopoietic / **blood and bone marrow** (UBERON:0000178 blood; UBERON:0002371 bone marrow). Body system: cardiovascular/hematologic (coagulation-hemostasis).
- **Primary cells: megakaryocytes** (CL:0000556) in bone marrow; **platelets** (CL:0000233) in circulation.
- **Secondary involvement:** mucosal/cutaneous bleeding sites (skin, nasal mucosa, endometrium) are affected by the bleeding tendency, not by intrinsic TPM4 pathology there.
- **Subcellular:** actin cytoskeleton (GO:0015629), stress fiber (GO:0001725), demarcation membrane system.
- **Lateralization:** not applicable (systemic hematologic disorder).

8. Temporal Development

- **Onset: Congenital/constitutional**; thrombocytopenia is lifelong and typically present from birth, though often detected incidentally later in life.
- **Onset pattern:** chronic/insidious (frequently asymptomatic until an incidental CBC or a bleeding challenge).

- **Progression: Stable, non-progressive.** No disease staging; platelet counts remain mild-to-moderately low throughout life (PMID: 35170221).
- **Duration:** chronic, lifelong.
- **Remission:** none expected (constitutional). No spontaneous resolution.
- **Critical periods:** hemostatic challenges (surgery, childbirth, trauma, dental procedures, menstruation) are the windows where intervention matters.

9. Inheritance and Population

- **Inheritance: Autosomal dominant** (heterozygous variants segregate with phenotype; PMID: 28134622, PMID: 35170221).
- **Penetrance:** appears high for the hematologic phenotype within reported pedigrees (e.g., 5 affected relatives in the Spanish family), but **expressivity of the bleeding tendency is variable** (mild/absent to excessive, variant-dependent).
- **Anticipation / germline mosaicism / founder effects:** none reported.
- **Consanguinity:** not relevant (dominant).
- **Epidemiology: Very rare**; only a handful of families reported worldwide (two BRIDGE-BPD families, one Spanish family, and small missense series). Precise prevalence/incidence **not established** — likely underdiagnosed and under-reported because of the mild phenotype and overlap with ITP.
- **Demographics:** no established ethnic predilection; reported in European pedigrees. Sex ratio not established; women may present more often clinically due to menorrhagia.

10. Diagnostics

- **Laboratory tests: Complete blood count** showing low platelet count (HP:0001873) with **increased mean platelet volume** and large platelets on **peripheral blood film review** — the key screening findings. Platelet **aggregation and secretion studies** may be **normal** (especially truncating variants) or mildly abnormal (missense variants) (PMID: 35170221, PMID: 34758189).
- **Specialized tests:** platelet spreading assays (impaired after TRAP-6/CRP) and immunofluorescence showing TPM4 delocalization in activated platelets (PMID: 35170221).
- **Genetic testing (definitive): Whole-exome sequencing** or **inherited platelet disorder / thrombocytopenia gene panels** including TPM4; single-gene testing confirms a familial variant. Genetic diagnosis is essential because the phenotype mimics other

macrothrombocytopenias and ITP (PMID: 19408193 for the inherited-platelet-disorder diagnostic paradigm).

- **Imaging / CMA / karyotype / FISH / mtDNA / repeat testing:** not indicated (no structural, mitochondrial, or repeat-expansion component).
- **Differential diagnosis:** immune thrombocytopenia (ITP) — key to avoid unnecessary immunosuppression/splenectomy; other inherited macrothrombocytopenias (MYH9-related disorder, Bernard–Soulier syndrome, GP1BA/B, ACTN1-related, TUBB1-related, ITGA2B/ITGB3-related); Glanzmann thrombasthenia and von Willebrand disease for bleeding work-up. Distinguishing features: TPM4 disorder is mild, lifelong, autosomal dominant, with large platelets and often-normal function.
- **Screening:** cascade **genetic testing of at-risk relatives** once a familial variant is identified; no population newborn screening.

11. Outcome / Prognosis

- **Survival/mortality: Normal life expectancy**; no excess mortality reported. Not a life-limiting condition.
- **Morbidity:** low overall; limited to bleeding events (notably menorrhagia and peri-procedural bleeding). Missense-variant carriers can have higher bleeding morbidity (PMID: 34758189).
- **Complications:** bleeding with surgery/trauma; iron-deficiency anemia from menorrhagia; risk of misdiagnosis as ITP leading to unnecessary immunosuppressive treatment or splenectomy.
- **Recovery/QoL:** generally excellent with appropriate recognition and supportive management.
- **Prognostic factors:** variant type (truncating → predominantly count defect, milder bleeding; missense → added platelet-function defect, potentially more bleeding) is the main prognostic determinant.

12. Treatment

There is **no disease-specific or curative therapy**; management is **supportive and bleeding-oriented**, extrapolated from general inherited-platelet-disorder care.

Intervention	Role	NCIT suggestion
Tranexamic acid (antifibrinolytic)	Mucosal bleeding, menorrhagia, peri-procedural	NCIT Tranexamic Acid; CHEBI:48669
Desmopressin (DDAVP)	May improve hemostasis for minor procedures	NCIT Desmopressin; CHEBI:4450
Hormonal therapy / combined oral contraceptives	Menorrhagia control	NCIT Oral Contraceptive
Platelet transfusion	Severe bleeding or major surgery (reserve; HLA considerations)	NCIT Platelet Transfusion
Recombinant activated factor VIIa	Refractory bleeding in inherited platelet disorders (adjunct)	NCIT Recombinant Factor VIIa
Iron supplementation	Treat iron-deficiency anemia from chronic bleeding	NCIT Iron Supplement

- **Pharmacogenomics / gene therapy / cell therapy / RNA therapy / targeted or immunotherapy: none applicable/available** for this disorder.
- **Treatment strategy:** avoid misdiagnosis-driven ITP therapy (steroids, IVIG, splenectomy); manage bleeding proactively around hemostatic challenges; involve hematology and genetic counseling. The general supportive toolkit for inherited platelet bleeding (antifibrinolytics, hormonal therapy, desmopressin, platelet support) parallels approaches used in other inherited platelet disorders.

13. Prevention

- **Primary prevention:** not possible (germline). **Genetic counseling** for autosomal dominant transmission (50% transmission risk); options include **prenatal testing** and **preimplantation genetic diagnosis** where a familial variant is known.
- **Secondary prevention:** cascade genetic testing of relatives; correct early diagnosis to prevent iatrogenic harm (unnecessary immunosuppression/splenectomy).
- **Tertiary prevention:** peri-procedural planning (antifibrinolytics, platelet support), menorrhagia control, iron repletion, and avoidance/caution with antiplatelet and anticoagulant drugs.
- **Immunization / public health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

- **Model species: Mouse** (*Mus musculus*, NCBI Taxon 10090); orthologous gene **Tpm4** (NCBI Gene 326618). ENU-induced *Tpm4* missense mutations and targeted *Tpm4* inactivation reproduce gene-dosage-dependent macrothrombocytopenia (PMID: 28134622).
- **Natural disease in companion animals/wildlife:** none reported (no OMIA entry identified).
- **Comparative biology / conservation:** the mechanism is **evolutionarily conserved** — *"a nonredundant role for TPM4 in platelet biogenesis in humans and mice"* (PMID: 28134622).
- **Zoonotic potential:** not applicable.

15. Model Organisms

- **Mouse models (primary):**
- **ENU-induced *Tpm4* missense mutants** — gene-dosage-dependent macrothrombocytopenia (PMID: 28134622).
- **Targeted *Tpm4* inactivation (knockout)** — macrothrombocytopenia with otherwise normal blood counts, modeling haploinsufficiency (PMID: 28134622).
- **Cellular/in vitro models:** human and mouse **megakaryocyte cultures** showing the terminal platelet-production defect; patient-derived platelets for spreading/secretion assays (PMID: 28134622, PMID: 35170221); **in vitro reconstituted actin–tropomyosin filaments** characterizing Tpm4.2 biochemistry (PMID: 28216317).
- **Phenotype recapitulation:** mouse models recapitulate the core macrothrombocytopenia (low count, enlarged platelets) with gene-dosage dependence — a strong match to human disease.
- **Limitations:** mild bleeding phenotype and subtle platelet-function defects are harder to model; missense dominant-negative effects may differ between species. Resource: **MGI** for *Tpm4* alleles.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
28134622	*Mutations in tropomyosin 4 underlie a rare form of human macrothrombocytopenia*	Human clinical + mouse + in vitro	Disease-defining: truncating TPM4 → AD macrothrombocytopenia; gene-dosage mechanism; terminal platelet-production defect; conserved human/mouse role
35170221	*A novel nonsense variant in TPM4 caused dominant macrothrombocytopenia...*	Human clinical (Spanish pedigree)	Independent confirmation: p.Gln108*; ↓TPM4 protein; impaired spreading; mild lifelong phenotype with normal aggregation/secretion
34758189	*Rare missense variants in TPM4 are associated with platelet dysfunction...excessive bleeding*	Human clinical	Missense end of spectrum: p.R182C, p.A183V; reduced secretion, aggregation/spreading defects, disordered tropomyosin; excessive bleeding
25964668	*Regulation of actin polymerization by tropomodulin-3 controls megakaryocyte actin organization and platelet biogenesis*	Mouse / cell biology	Places TPM4/TM4 in the TMOD3–F-actin module of proplatelet formation
28216317	*Tropomyosin isoforms specify functionally distinct actin filament populations in vitro*	In vitro biochemistry	Tpm4.2 binds F-actin with high affinity, competes with α-actinin — molecular basis for destabilization
28101865	*Tropomyosin structure, function, and interactions: a dynamic regulator*	Review	Tropomyosin as universal specifier/regulator of the actin cytoskeleton
19408193	*Genetic testing in the diagnostic evaluation of inherited platelet disorders*	Review	Diagnostic paradigm for inherited platelet disorders (context)

Supporting context literature: cytoskeletal tropomyosin structure/function (PMID: 36586407, PMID: 32037094, PMID: 10986121); general inherited-platelet-disorder management (PMID: 41201390, PMID: 40932452). Unrelated TPM4 cancer biology (not disease-causal here): PMID: 40214825, PMID: 31804537.

Limitations and Knowledge Gaps

1. **Very small clinical evidence base:** only ~3 pedigrees/series directly define the disorder (two BRIDGE-BPD families, one Spanish family, a small missense series). Prevalence, penetrance, and expressivity estimates are therefore uncertain.
2. **Incomplete formal nosology:** OMIM, Orphanet, ICD, and MeSH identifiers were not confirmed in the available evidence; standardized synonyms are not catalogued.
3. **Variant-level data gaps:** exact gnomAD frequencies, full ACMG/AMP classification detail, and the complete truncating-variant coordinates from the 2017 families were not captured here.
4. **Mechanistic inference:** the demarcation-membrane / proplatelet-architecture step (step 3 of the causal chain) is inferred from the TMOD3–TM4–F-actin module rather than directly demonstrated for pathogenic TPM4 variants.
5. **Missense mechanism uncertainty:** whether missense alleles act by simple loss of function, dominant-negative mis-specification, or both is not fully resolved.
6. **No treatment trials:** all management is extrapolated/supportive; there are no disease-specific therapeutic studies or clinical-trial data (no NCT identifiers).

Proposed Follow-up Experiments / Actions

1. **International variant/case registry:** aggregate all reported and new TPM4 families to estimate prevalence, penetrance, genotype–phenotype correlation (truncating vs missense), and bleeding scores (ISTH-BAT).
2. **Formal nosology curation:** confirm/create OMIM, Orphanet, and ICD-11 entries and a canonical synonym set; reconcile with MONDO:0100487.

3. **Mechanistic validation of step 3:** live-imaging of proplatelet formation and demarcation-membrane dynamics in patient-derived iPSC megakaryocytes carrying truncating vs missense TPM4 variants.
4. **Allele-mechanism dissection:** biochemical and structural (cryo-EM) comparison of wild-type vs p.R182C/p.A183V Tpm4.2 on actin filaments to test dominant-negative vs loss-of-function.
5. **Diagnostic guidance:** promote inclusion of TPM4 on thrombocytopenia panels and develop a simple clinical algorithm (large platelets + AD inheritance + often-normal aggregation → sequence TPM4) to reduce ITP misdiagnosis.
6. **Natural-history and bleeding-management study:** prospective assessment of peri-procedural outcomes and the efficacy of antifibrinolytics/desmopressin/hormonal therapy specifically in TPM4 carriers.

Evidence-source legend: human clinical = patient/pedigree studies; model organism = mouse; in vitro = biochemistry/cell biology; review = synthesis. All mechanistic and clinical claims are cited to the PMIDs above; inferred steps are labeled as such.
